Asset Performance:

Hea 08 Ventilation system air intakes and exhausts

2 Credits

0

No Minimum Standard

Aim

To ensure that the asset ventilation system minimises the entry of external sources of air pollution.

Question

Are ventilation system air intakes and exhausts located to minimise the entry of air pollutants into the asset?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	No
2	C.	Yes

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	For assets with air-conditioned or mixed-mode ventilation systems: EITHER a) The location of the asset's air intakes and exhausts, in relation to each other and to sources of external pollution, are in accordance with either of the following standards: • Sections 8.8.2-8.8.4 of PD CEN/TR 16798-4:2017 Energy performance of buildings - Ventilation for buildings - Part 4: Interpretation of the requirements in EN 16798-3 - For non-residential buildings - Performance requirements for ventilation and room-conditioning systems (Modules M5-1, M5-4) • Section 5.5 and Normative Appendix B of ANSI/ASHRAE Standard 62.1-2019 Ventilation for Acceptable Indoor Air Quality. Alternatively, where local standards have similar requirements to any of the above standards, the local standard may be used to demonstrate compliance with this criterion. OR	C

Criterion	Assessment criteria	Applicable Answer
	b) The asset's air intakes are positioned at least 10m horizontal distance from sources of external pollution, including the location of air exhausts from the asset and other buildings. Exhausts or other pollutant sources are not to be discharged into enclosed external spaces, such as courtyards, in which intakes are also located.	
2.	For naturally ventilated assets, background ventilators and openable windows, skylights and doors are positioned at least 10m horizontal distance from sources of external pollution, including the location of air exhausts from other buildings. Exhausts or other pollutant sources are not to be discharged into enclosed external spaces, such as courtyards, in which intakes are also located.	C

Methodology

Pollutant dispersion modelling

If compliance cannot be demonstrated using the methods stipulated in the assessment criteria, it may be feasible to demonstrate compliance using pollutant dispersion modelling if this shows that, under normal asset operational and typical external environmental conditions, the location of the asset's air intakes and exhausts in relation to each other and sources of external pollution minimises the entry of air pollutants into the asset. This can be achieved using either wind tunnel modelling or numerical modelling. Pollutant dispersion modelling in urban areas is complex, so it is important that the person carrying out the modelling is a competent individual.

Jet exhaust systems

Jet exhaust systems may be used as an alternative method of demonstrating compliance with criterion 1 if evidence from pollutant dispersion modelling demonstrates that the location and separation of air intakes and the jet exhausts, the jet exhaust nozzle direction, and the average jet exhaust velocity prevents significant recirculation of exhaust air under typical wind conditions.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Documentation (e.g. plans, photographs or scaled drawings of the asset) showing the location of air intakes and exhausts and external sources of pollution and the distances between them.

Definitions

Competent individual – wind tunnel modelling:

An individual with the following qualifications and experience:

- Holds a degree or equivalent qualification in a relevant engineering field (mechanical, chemical), physics, mathematics, or meteorology AND holds a membership of an appropriate professional body.

- Has a minimum of three years relevant experience, which must clearly demonstrate a practical understanding and experience of wind tunnel modelling and factors affecting outdoor pollutant dispersion in relation to ventilation and the built environment.

Competent individual – numerical modelling:

An individual with one or more of the following qualifications and experience:

- Holds a degree or equivalent qualification in a relevant engineering field (mechanical, chemical), physics, mathematics, meteorology, environmental sciences, environmental engineering or a related environmental discipline, AND holds a membership of an appropriate professional body,
- Demonstrable ability to interpret environmental guidelines, policies, plans and legislative requirements.

Numerical modelling:

A computer-based stimulation method for modelling pollutant dispersion and air quality in the outdoor environment. Various numerical models are commercially available which may be used to investigate the location of ventilation intakes and exhausts, including those based on empirical methods and computational fluid dynamics (CFD).

Sources of external pollution:

These include but are not limited to the following:

- Highways and the main access roads associated with the asset
- Car parks, delivery and vehicle pick-up, drop-off or waiting bays
- Other building-related emissions, including from building services exhausts, industrial or agricultural processes, and external smoking areas.

Service and access roads with restricted and infrequent access, e.g. roads used only for waste collection, are unlikely to represent a significant source of external pollution. These roads can therefore be excluded from assessment.

Wind tunnel modelling:

This is a versatile physical technique which allows many variables (for example building design, intake and exhaust positions, local pollutant sources, wind speed and direction), to be investigated for complex urban areas. Wind tunnel modelling provides reliable and detailed data, both visual and quantitative, on outdoor pollution distribution. This enables the effective siting of intakes and exhausts to be evaluated for both mechanically and naturally ventilated buildings.